Human Development Index (HDI) Predictor
Solution Requirements

Project: Human Development Index (HDI) Predictor
Phase: Requirement Analysis

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1. Functional Requirements

- Accept a country name plus 4 numeric indicators (Life Expectancy, Mean Years of Schooling, GNI per Capita, Internet Users)
- Return a predicted HDI score and a Low/Medium/High/Very High development category
- Provide a Home page explaining the HDI and linking to the Predict page
- Handle invalid or out-of-range input without crashing the application

2. Non-Functional Requirements

- Response time under a few hundred milliseconds on the Flask development server
- Codebase split into clear, reusable modules (Training/ for the ML pipeline, Flask/ for the web app)
- Model should generalize well: target R^2 above 0.90 on held-out test data
- App should run correctly regardless of the working directory it is launched from

3. Data Requirements

- A dataset with Country, Life Expectancy, Mean Years of Schooling, GNI per Capita, Internet Users, and HDI
- Coverage across the full development spectrum (Low, Medium, High, Very High) for the model to generalize
- Missing values handled via column-mean imputation before training

4. Constraints

- Model must remain lightweight and interpretable (Linear Regression, not a black-box model)
- Web app runs locally with the Flask development server for this internship submission